Printed Pages: 02 Sub Code: KEC074

Paper Id: 2 3 1 9 1 2 Roll No.

B.TECH.
(SEM VII) THEORY EXAMINATION 2022-23
MICROWAVE & RADAR ENGINEERING

Time: 3 Hours *Total Marks: 100*

Note: Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. **Attempt *all* questions in brief.** 2x10 = 20

(a) Explain various losses present in waveguides.
(b) Define characteristic impedance.
(c) Define the term directivity and coupling factor in directional coupler.
(d) Write down the different properties of s-matrix.
(e) Write down the limitations of conventional tubes at microwave frequencies.
(f) Explain why slow wave structure is used in TWTs.
(g) Define the term VSWR.
(h) Define the term Dielectric Constant.
(i) Discuss the concept of low angle tracking used in RADAR.
(j) Define the term pulse repetition frequency in radar.

SECTION B

2. **Attempt any *three* of the following:** 10x3 = 30

(a) Derive the field distribution of TE10 mode in rectangular waveguide. Show that TE01 and TM10 modes do not exist in rectangular waveguide.
(b) Explain the operation of a Faraday rotation isolator with the help of neat sketch. List the applications of ferrite isolator.
(c) Compare TWT and Klystron amplifier. Explain the amplification process of Travelling Wave Tubes (TWT).
(d) Discuss the salient features of microwave measurements. Describe a voltage standing wave ratio (VSWR) meter.
(e) Explain with neat diagram, the working of CW radar with non zero IF.

SECTION C

3. **Attempt any *one* part of the following:** 10x1 = 10

(a) Derive the expression for field components in TE mode for circular waveguide.
(b) TE11 wave is propagating through a circular waveguide. The waveguide is air filled and has a diameter of 12cm. Calculate cut-off frequency (fc), guide wavelength for frequency of 2.5 GHz (λg) and wave impedance (Zg).

Microwave & Radar

(2)

4. **Attempt any *one* part of the following:** 10 x 1 = 10

(a) With a schematic describe the operation of a four port circulator. Obtain the simplified S matrix of a perfectly matched, lossless four port circulator.

(b) Write short note on:

i. Phase Shifters

ii. Attenuators

5. **Attempt any *one* part of the following:** 10x1 = 10

(a) Explain the schematic diagram, operating principle and performance characteristics of Backward Wave Oscillators (BWO).

(b) A Two-CavityKlystron amplifier has the following parameters: V_0 = 1000 V, R_0 = 40 kilo-ohm, I_0 = 25 mA, f = 3 GHz, d = 1 mm, L = 4 cm, R_{sh} (effective shunt impedance, excluding beam loading) = 30 kilo-ohm.

i. Find the input gap voltage to give maximum voltage V2.

ii. Find the voltage gain, neglecting the beam loading in the output cavity.

iii. Find the efficiency of the amplifier, neglecting beam loading.

6. **Attempt any *one* part of the following:** 10x1 = 10

(a) Write short note on:

i) Power Meters

ii) Microwave Amplifiers

(b) Write short note on:

i) Measurement of Insertion Loss

ii) ReturnLoss

7. **Attempt any *one* part of the following:** 10x1 = 10

(a) Derive simple form of radar range equation.

(b) A typical pulse radar waveform has T_{on} =10μs, pulse repetition period (T_r) = 10msand peak power= 10 MW.

Find:

i) Duty cycle

ii) Maximum unambiguous range (R_{un})

iii) Average power (P_{avg})

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Subject Code: KEC503

Roll No:

BTECH
(SEM V) THEORY EXAMINATION 2023-24
DIGITAL SIGNAL PROCESSING

TIME: 3 HRS   M.MARKS: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.   **2 x 10 = 20**

Q no.	Question	Marks	CO
a.	Distinguish between IIR and FIR filter.	2	1
b.	Define linear phase response of a filter.	2	1
c.	Compare bilinear transformation and Impulse invariant method of IIR filter design.	2	2
d.	Distinguish between Butterworth and Chebyshev (Type-I) filter.	2	2
e.	State the need for employing window for designing FIR filter?	2	3
f.	The most straight forward approach to filter design is to truncate the impulse response of an ideal IIR filter. Why this is usually an undesirable approach?	2	3
g.	Develop the 4-point DFT of the sequence $x(n) = \{1,1\}$.	2	4
h.	How is the FFT algorithm algorithm applied to determine inverse discreate Fourier transform?	2	4
i.	Highlight the features of a commercial digital signal processor.	2	5
j.	Explain the concept of multistage sampling rate conversion.	2	5

SECTION B

2. Attempt any *three* of the following: **10x3=30**

a.	Determine the cascade and parallel realization for the system transfer function $$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$	10	1
b.	Use bilinear transformation to convert low pass filter $$H(s) = \frac{1}{(1 + 1.41s + s^2)}$$ into a high pass filter with pass band edge at 100 Hz and Fs=1 kHz.	10	2
c.	Design a low pass FIR filter for the following specifications using rectangular window function Cut-off frequency = 500 Hz; Sampling frequency = 2000 Hz; Order of the filter = 10	10	3
d.	The first five points of the 8-point DFT of a real valued sequence are {0.25, 0.125-j0.3018, 0, 0.12-j0.0518, 0}. Determine the remaining three points.	10	4
e.	Explain the various types of addressing modes of digital signal processor with suitable example.	10	5

SECTION C

3. Attempt any *one* part of the following: **10x1=10**

a.	Realize the given H(z) for using ladder structure $$H(z) = \frac{2 + 8z^{-1} + 6z^{-2}}{1 + 8z^{-1} + 12z^{-2}}$$	10	1

Subject Code: KEC503

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BTECH
(SEM V) THEORY EXAMINATION 2023-24
DIGITAL SIGNAL PROCESSING

TIME: 3 HRS M.MARKS: 100

b.	Obtain the direct form-I and direct form-II realization of a given LTI system: $y(n) = -0.1y(n-1) - 0.72y(n-2) - 0.7x(n) - 0.25x(n-2)$	10	1

4. Attempt any *one* part of the following: 10x1=10

a.	Convert the analog filter with system function $$H_a(s) = \frac{(s+0.1)}{(s+0.1)^2+9}$$ into a digital IIR filter by means of the impulsive invariance method.	10	2
b.	Design a Chebyshev filter for the following specification using bilinear transformation. $0.8 \le \lvert He^{jw}\rvert \le 1 \quad 0 \le w \le 0.2\pi$; $\lvert He^{jw}\rvert \le 0.2 \quad 0.6\pi \le w \le \pi$	10	2

5. Attempt any *one* part of the following: 10x1=10

a.	What is Hamming Window Function? Obtain its frequency domain characteristics.	10	3
b.	Design a low pass filter using Kaiser window satisfying the specifications given filter: Passband cutoff frequency F_p = 150 Hz Stopband cutoff frequency F_s = 250 Hz Sampling frequency F_t = 1000 Hz Passband attenuation A_p = 0.1dB Stopband attenuation A_s = 40 dB	10	3

6. Attempt any *one* part of the following: 10x1=10

a.	Determine the DFT of the given sequence x(n) = {1,2,3,4,4,3,2,1} using DIF FFT algorithm.	10	4
b.	Find the output y(n) of a filter whose impulse response is h(n) = {1,1,1} and input signal x(n) = {3,-1,0,1,3,2,0,1,2,1} using overlap add method.	10	4

7. Attempt any *one* part of the following: 10x1=10

a.	Explain the process of multirate signal processing in detail. Also, enlist the advantages of multirate signal processing.	10	5
b.	Write the short note on: (i) Recursive Least Square Algorithm (ii) (ii) Window LMS Algorithm	10	5

PAPER ID-310581

Subject Code: KEC076

Roll No:

BTECH
(SEM VII) THEORY EXAMINATION 2023-24
WIRELESS & MOBILE COMMUNICATION

TIME: 3 HRS | M.MARKS: 100

Note: 1. Attempt all Sections. If require any missing data, then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

Qno.	Question	Marks	CO
a.	Illustrate the need to evolve third generation wireless standards.	2	1
b.	Write the name of channel assignment strategies in mobile radio propagation.	2	1
c.	Discuss characteristics of a pn sequence.	2	2
d.	Define the term vocoder.	2	2
e.	Illustrate the maximum throughput efficiency of Slotted ALOHA.	2	3
f.	Define equalization.	2	3
g.	Illustrate the terms SGSN and GGSN.	2	4
h.	Illustrate the function of Node B in UMTS.	2	4
i.	Explain the advantage of light fidelity.	2	5
j.	Discuss Wi-Fi communication in a cellular system.	2	5

SECTION B

2. Attempt any *three* of the following:

a.	Explain co-channel reuse ratio. Also derive the relationship between co-channel reuse ratio and cluster size.	10	1
b.	Illustrate slow FHSS and fast FHSS in detail.	10	2
c.	Illustrate CSMA and CSMA/CD with the help of proper flow diagram.	10	3
d.	Explain GPRS architecture in detail. Also explain the use of CCU and PCU unit in detail.	10	4
e.	Illustrate the advantage of NGN networks.	10	5

SECTION C

3. Attempt any *one* part of the following:

a.	If a signal to interference ration of 17 dBm is required for satisfactory forward channel performance of a cellular system. Calculate the frequency reuse factor and cluster size that should be used for maximum capacity if the path loss exponent (n) is 4. Assume that there are 6 co-channel cells in first tier and all of them are at the same distance from the mobile. Use suitable approximations.	10	1
b.	Explain Frequency Reuse concept with the help of proper cellular diagram. Also explain umbrella cell concept in mobile communication.	10	1

4. Attempt any *one* part of the following:

a.	Explain various types of vocoders with brief view of general voice generation mechanism.	10	2
b.	Explain various diversity techniques in wireless and mobile communication.	10	2

5. Attempt any *one* part of the following:

a.	Illustrate various equalization techniques with the help of proper block diagram.	10	3
b.	Compare FDMA, CDMA and TDMA in detail.	10	3

6. Attempt any *one* part of the following:

a.	Explain UMTS architecture in detail. Also briefly explain IMT 2000.	10	4
b.	Compare LEO, MEO, GEO. Also explain the timing diagram of call connection between two ISU units in mobile satellite communication.	10	4

7. Attempt any *one* part of the following:

a.	Discuss Mobile Ad-hoc Network (MANET) in detail.	10	5
b.	Write short note on (i) Wi-Max (ii) 4g technology	10	5